Machine Learning-Based Protein Biomarker Discovery for Rheumatoid Arthritis Using Olink Proteomics

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Abstract

Rheumatoid arthritis (RA) is a chronic autoimmune disease characterized by persistent synovial inflammation, joint destruction, and substantial clinical heterogeneity. Current diagnostic strategies, based on clinical assessment and serological markers such as rheumatoid factors (RFs) and anti-citrullinated protein antibodies (ACPAs), are insufficient to capture disease heterogeneity, particullarly in seronegative patients. High-throughput proteomics offers the opportunity to identify circulating biomarkers that reflect underlying pathogenic mechanisms, thereby enabling improved disease stratification. In this project, Olink Explore HT proteomics data were analyzed using machine learning approaches to identify protein biomarkers that classify RA subdiagnoses. In particular, multinomial elastic net (GLMnet) models were trained to distinguish healthy controls from RA subtypes, with additional models incorporating age and sex as features, and sex-stratified analyses. Model performance was evaluated using cross-validation, and important features were extracted based on model coefficients. Several biologically relevant proteins, including PDCD1, IL6, PADI4, CXCL13, and MSMP, were consistently identified as key contributors to classification. Functional enrichment analysis further supported the involvement of immune activation, T-cell regulation, and B-cell-mediated responses. Together, these results demonstrate that the machine learning model trained on large-scale proteomics data can identify biologically meaningful biomarkers with potential diagnostic, therapeutic and stratification value in RA.

Introduction

RA is a chronic, systemic autoimmune disease characterized by persistent synovial inflammation, progressive joint destruction, and frequent extra-articular involvement. RA leads to irreversible disability and long-term impairment in physical function and quality of life if insufficiently treated (Smolen et al. 2016). Despite major therapeutic advances over the past two decades, RA remains incurable, with a large proportion of patients failing to achieve sustained remission.

As one of the most prevalent systemic autoimmune diseases worldwide, RA has an estimated global prevalence of approximately 0.3-1.0%, with higher prevalence reported in industrialized regions, such as North America and Western Europe (Smolen et al. 2016; Finckh et al. 2022). RA incidence increases with age and displays an obvious sex bias. Women are affected approximately two- to three-fold more frequently than men, with an estimated lifetime risk of about 3.6% for women and 1.7% for men. In addition to joint disease, RA is associated with CVD, which reflects its systemic nature and complicated mechanisms (Smolen et al. 2016).

The development of RA is related to both genetic and environmental factors. Twin studies suggest a heritability of approximately 60% for ACPA-positive RA, while seronegative RA shows a lower genetic contribution (Smolen et al. 2016). The strongest

genetic associations map to the Human Leukocyte Antigen (HLA) class II region, especially HLA-DRB1 alleles that encode the shared epitope, which influences antigen presentation and adaptive immune activation. Beyond HLA, genome-wide association studies (GWAS) have identified over 100 susceptibility loci, most of which are enriched in immune and inflammatory pathways rather than protein-coding regions (Smolen et al. 2016). In terms of environmental factors that trigger disease in genetically susceptible individuals, smoking is the most consistently established risk factor and shows a strong interaction with HLA-DRB1 shared epitope alleles, particularly in ACPA-positive RA (Smolen et al. 2016). Additional environmental contributors include silica exposure, obesity, hormonal factors, and alterations in the microbiota.

RA pathogenesis develops through several stages and often begins years before clinical onset. In the preclinical phase, post-translationally modified self-proteins, such as citrullinated proteins, trigger adaptive immune responses and autoantibody production, including ACPAs and RFs (Smolen et al. 2016; Ding et al. 2023). These autoantibodies can be detected long before the clinical appearance of apparent synovitis. Progression to clinical RA is typically marked by synovial immune cell infiltration and activation of resident stromal cells. Fibroblast-like synoviocytes (FLS) acquire an aggressive, apoptosis-resistant phenotype and play a central role in pannus formation and joint destruction (Ding et al. 2023). Both innate and adaptive immune cells, including macrophages, dendritic cells, T cells, and B cells, accumulate within the synovium and form a self-sustaining inflammatory microenvironment.

In addition, cytokine and chemokine networks are key drivers of disease persistence and tissue damage. Central mediators include tumor necrosis factor (TNF), interleukin-6 (IL6), interleukin-1 (IL1), and granulocyte-macrophage colony-stimulating factor (GM-CSF) (Smolen et al. 2016; Ding et al. 2023). These cytokines activate downstream signaling pathways, such as JAK–STAT, NF-κB, MAPK, and PI3K–AKT, and lead to immune cell activation, synovial hyperplasia, angiogenesis, and osteoclast-mediated bone erosion (Ding et al. 2023). Importantly, RA pathogenic mechanisms are not uniform across all patients. ACPA-positive and ACPA-negative RA differ in genetic associations, immune activation patterns, and clinical appearances, suggesting the heterogeneous nature of RA (Smolen et al. 2016).

RA diagnosis is based on several clinical evaluation strategies, including serology, imaging, and classification criteria such as those developed by the American College of Rheumatology and the European League Against Rheumatism (Ding et al. 2023). While RF and ACPAs improve diagnostic specificity, a large fraction of patients is seronegative, making it difficult to distinguish RA subtypes. The first-line treatment strategy uses disease-modifying antirheumatic drugs (DMARDs). Biologic and targeted synthetic DMARDs targeting TNF, IL-6 signaling, T-cell co-stimulation, B cells, and JAK-STAT pathways have markedly improved outcomes for many patients (Smolen et al. 2016; Ding et al., 2023). However, treatment response remains highly variable among patients of different backgrounds, and many patients experience

inadequate response and adverse effects.

A research gap appears as current diagnostic markers are insufficient to differentiate RA subtypes accurately. Therapeutic strategies also have a limited ability to provide patients with the most effective targeted therapy based on underlying pathogenic mechanisms (Smolen et al. 2016). Therefore, there is a critical need for biomarkers that enable RA subtype stratification and support mechanism-informed diagnosis and therapy.

Circulating protein biomarkers that capture dynamic disease states and pathway activity are well-suited for disease stratification and pathogenic mechanism explanation. Recent advances in affinity-based proteomics have significantly improved plasma protein profiling. Proximity extension assay (PEA)-based platforms enable sensitive and specific quantification of thousands of proteins from small plasma volumes. The integration of PEA with next-generation sequencing (NGS) further increases throughput and scalability for the profiling of large cohorts (Wik et al. 2021). Large population-scale studies using these platforms have shown stable individual proteomic signatures and disease-associated deviations in different conditions (Álvarez et al. 2025). Together, these developments provide a technical foundation to identify circulating protein signatures associated with RA subtypes and underlying pathogenic mechanisms.

Analytical methods of high-dimensional proteomics data include traditional univariate analyses that fail to model correlated proteins, and machine learning methods that enable feature integration and selection. Regularized regression model, a machine learning method, is well-suited for proteomics data due to its automatic feature selection ability. In this context, leveraging high-throughput plasma proteomics with machine learning methods offers a comprehensive framework that captures multivariate protein patterns linked to RA heterogeneity to address current diagnostic and therapeutic limitations in RA.

Materials and Methods

Study Cohort

Plasma proteomics data were collected from Human Disease Blood Atlas (https://disease.proteinatlas.org/) and proved for analysis in November 2025. The data were generated using the Olink Explore HT platform and measure 5,415 proteins per sample. The dataset included healthy controls and RA patients with annotated subdiagnoses as ACPA-positive and ACPA-negative (**Fig. 1**). Particularly, 50 healthy samples, 100 ACPA-positive samples and 100 ACPA-negative samples were included. Demographic variables, including age and sex, were available for all samples. Female patients are predominant in this sample group, which aligns with the fact that females are more likely to develop RA (Smolen et al. 2016).

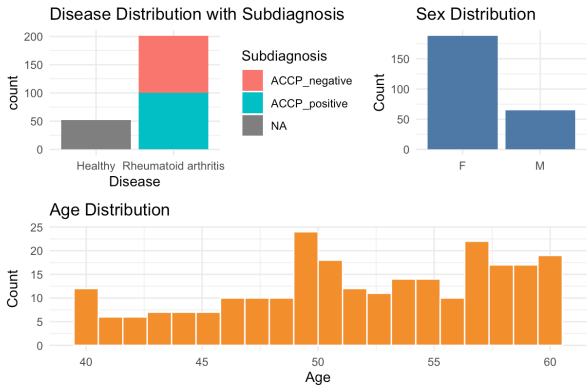

Fig. 1: Data distribution of RA samples.

Olink Proteomics

The Olink Explore HT assay is based on PEA combined with NGS. Briefly, once pairs of oligonucleotide-labeled antibodies bind to target proteins, they are extended to form unique DNA sequences, which are then amplified and quantified by NGS. Protein abundance is reported as Normalized Protein eXpression (NPX) values, which are relative expression values of targeted proteins generated by normalization, background correction and $\log_2$ transformation of raw NGS readouts. Higher NPX values represent higher a relative abundance of proteins.

Data Preprocessing

Protein expression values were already standardized prior to modeling. Among all samples, two samples (DA19171 and DA19162) with missing protein expression levels were excluded to ensure consistent input dimensionality across models. Disease status and subdiagnoses were encoded as categorical outcomes. Age was treated as a continuous variable and sex as a categorical variable when included as covariates. For all models, protein expression values were further standardized using z-score normalization to make them comparable using recipe-based preprocessing within the training set.

Model Framework

All predictive modeling was conducted using the *tidymodels* framework in R. Multinomial and binary classification models were implemented using generalized linear models with elastic net regularization using the *glmnet* package in R. Model hyperparameters included the regularization strength (penalty, λ) and the elastic net mixing parameter (mixture, α). Hyperparameter tuning was performed using grid search over a regular grid spanning penalty values from 10^{-5} to 10^{1} and mixture values from 0 (ridge) to 1 (lasso).

In total, four model settings were evaluated: (i. *Model 0*) RA versus healthy controls using protein expression only; (ii. *Model 1*) RA subdiagnosis versus healthy using protein expression only; (iii. *Model 2*) RA subdiagnosis classification including age and sex as covariates; and (iv. *Model 3*) sex-stratified RA subdiagnosis model.

Data were split into training and test sets using stratified sampling. Model training was performed using 5-fold cross-validation and elastic net hyperparameters were tuned within cross-validation. Model performances were assessed using multiclass ROC AUC and accuracy. Proteins were ranked based on absolute model coefficients and top-ranked proteins were extracted. For selected top-ranked proteins, biological expression patterns were examined by visualizing normalized protein abundance across subdiagnosis groups.

Functional Enrichment Analysis

To assess biological relevance, functional enrichment analysis was performed on the top-ranked proteins from *Model 1*. Protein symbols were mapped to Entrez gene identifiers using the *org.Hs.eg.db* annotation database. Gene Ontology (GO) biological process enrichment was conducted using the *clusterProfiler* package in R, with the full set of measured proteins used as the background universe. Enriched pathways were identified using a Benjamini-Hochberg adjusted p-value threshold of 0.05.

Biological Interpretation

For each top protein selected by the model, functional information was first retrieved from *UniProt* (https://www.uniprot.org/uniprotkb/), including protein summary information and known biological functions. Protein expression patterns were then examined by comparing with blood- and tissue-related data on *Human Protein Atlas* (HPA; https://www.proteinatlas.org/). Specifically, reported expression trends in RA and other inflammatory conditions were compared with the direction of effect reported by the model to assess consistency. In addition, a targeted literature review was conducted for each top protein to identify previously reported associations with RA, including association with disease activity, autoantibody status, immune cell involvement, and therapeutic choices and responses. This step was used to connect statistical feature selection with existing biological knowledge and potential mechanistic relevance.

Results

Model Performance

Overall, proteomics-based models showed good discriminative performance across classification tasks (**Supplementary Fig. 1**). *Model 0* achieved an ROC AUC of 0.842 on the held-out test set, indicating strong separation between disease and control samples. For the multiclass classification in *Model 1*, elastic net models were able to differentiate healthy controls, ACPA-positive RA, and ACPA-negative RA with consistent performance across classes. One-vs-rest ROC curves showed comparable AUC values of 0.888 for each class, suggesting balanced classification performance without dominance by a single group. Incorporation of age and sex as additional covariates (*Model 2*) resulted in similar multiclass performance, with ROC AUC values remaining at 0.888 across all classes. This indicated that demographic variables age and

sex provided limited improvement over proteomic features alone, and that subtype discrimination was primarily driven by circulating protein profiles. Sex-stratified models (*Model 3*) showed decreased performance but remained informative. In female samples, multiclass ROC AUC values were 0.767, while male-specific models achieved lower AUC values of 0.692. The decrease in performance was consistent with reduced sample size after stratification and suggested lower model stability in sex-specific analyses. Nevertheless, both models retained discriminative capacity above random expectation. Taken together, these results showed that high-dimensional plasma proteomics enabled robust classification of RA and its subdiagnoses. Model performance was stable across modeling strategies, with the strongest discrimination observed in non-stratified analyses.

Top Protein Selection

(i) *Model 0*

Model 0 aimed to distinguish RA patients from healthy controls using plasma proteomic features only. Feature selection resulted in a small set of proteins with non-zero coefficients and large effect sizes (**Supplementary Fig. 2**). Among the top 5 selected proteins, CXCL13, IL6, and TIMD4 showed positive coefficients for the RA class and negative coefficients for healthy controls, indicating higher expression levels in RA. In contrast, PLRG1 showed a negative coefficient for RA and a positive coefficient for healthy controls, suggesting relative enrichment in non-disease samples.

(ii) *Model 1*

Model 1 extended the analysis to multiclass classification of healthy controls, ACPA-positive RA, and ACPA-negative RA. Distinct protein contribution patterns were observed across disease subgroups (**Fig. 2A**). PDCD1 showed a strong positive association with ACPA-positive RA, while IL6 contributed positively to both RA subtypes relative to healthy controls. PADI4 exhibited a subtype-specific association, with positive coefficients for ACPA-positive RA and minimal contribution to other classes. Several proteins showed preferential association with ACPA-negative RA. These included FAM104B, NOS2, CCNL2, and MSMP, which displayed positive coefficients for ACPA-negative disease and negative coefficients for other classes. In contrast, KLK8 and CFHR5 were more strongly associated with the healthy control class. The results from this model were analyzed mainly in this study, as it achieved the major objective by classifying three groups purely based on proteomics data.

(iii) *Model 2*

Model 2 incorporated age and sex as additional covariates with proteomic features to assess their impact on subtype classification. The top selected proteins in *Model 2* largely overlapped with those identified in *Model 1*, including PDCD1, IL6, CXCL13, NOS3, and SERPINA9 (**Fig. 2B**). However, age and sex showed smaller coefficient magnitudes relative to proteomic predictors and did not substantially affect protein coefficients. Notably, MSMP remained strongly associated with ACPA-negative RA after adjustment, while PDCD1 and IL6 retained strong positive associations with

ACPA-positive disease. These findings again suggested that demographic variables provided limited additional discriminatory power beyond proteomic data.

(iv) *Model* 3

To explore potential sex-specific proteomic signatures, *Model 3* was trained separately in female and male subsets using protein features only. In female samples, the top-ranked proteins were primarily associated with the ACPA-positive RA class. SERPINA9, PDCD1, CXCL13, MZB1, and IL6 showed positive coefficients for ACPA-positive disease. In contrast, IL9, TNN, and SRBD1 were associated with the ACPA-negative class, while HBZ showed a weaker contribution (**Fig. 2C**). In male samples, a distinct set of top-ranked proteins was observed. RNASET2, TNFRSF10B, ACTN4, and CBS were strongly associated with ACPA-negative RA, whereas SAA4, SCGB3A1, and NOS3 showed positive coefficients for ACPA-positive disease. Proteins associated with the healthy control class, including MIF4GD and CT83, exhibited opposite coefficient directions relative to RA-associated features (**Fig. 2D**). Female- and male-specific protein sets had limited overlap, but the small sample size of the male group limited the ability of the model to identify true sex-dependent proteomic signatures.

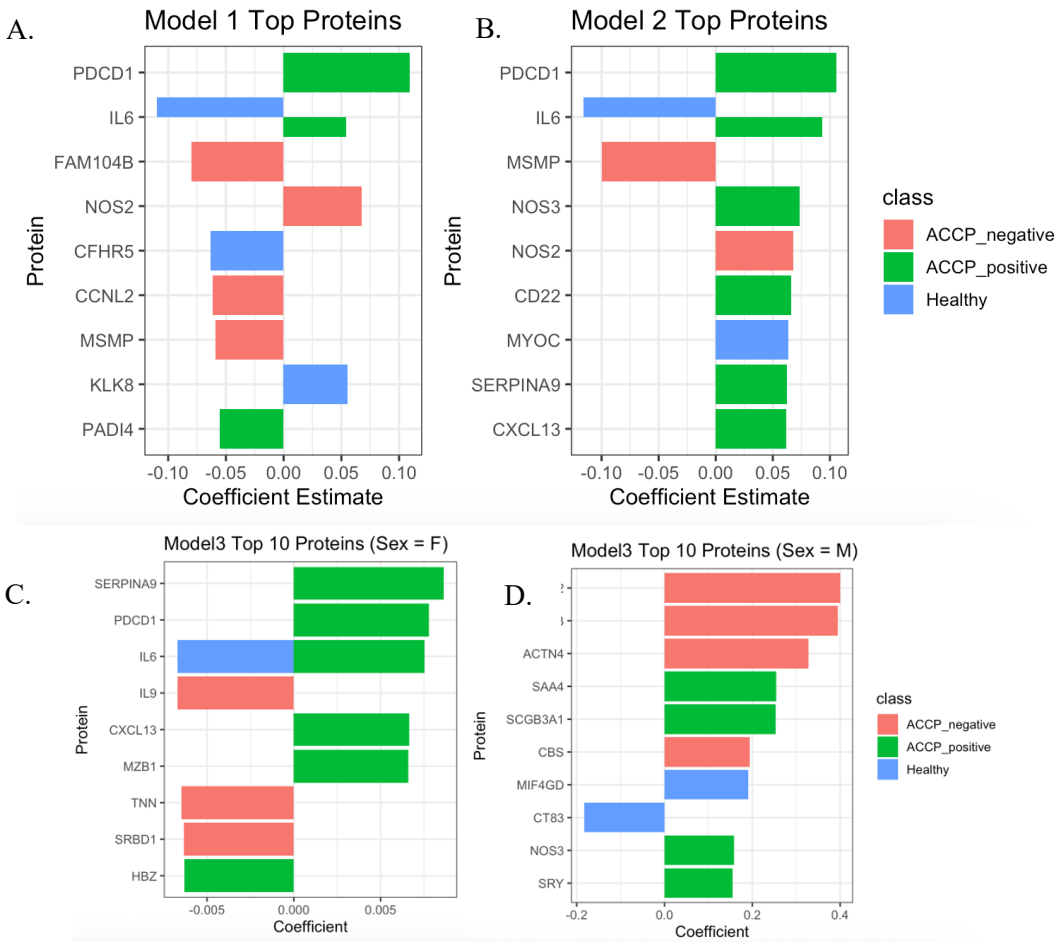

Fig. 2: Top-ranked proteins selected by *Model 1* (A), *Model 2* (B), *Model 3* (C) and *Model 4* (D).

Expression Pattern of Top Selected Proteins

Protein expression levels were further compared across subdiagnosis groups for the top selected proteins of *Model 1*. Clear and structured expression differences were observed between ACPA-negative RA, ACPA-positive RA, and healthy controls (**Fig 3**). Several proteins showed consistently higher expression in RA compared with healthy controls. IL6 displayed markedly elevated NPX values in both RA subtypes, with the highest median expression observed in ACPA-positive disease. A similar pattern was observed for CXCL13, which showed progressive upregulation from healthy controls to ACPA-negative RA and further to ACPA-positive RA. CD22 and MZB1 showed higher expression in ACPA-positive RA compared with ACPA-negative RA and healthy controls. The separation between ACPA-positive disease and the other groups was also obvious for MZB1. Several proteins exhibited reduced expression in RA relative to healthy controls. KLK7, MYOC, and SCGB1A1 showed higher NPX values in healthy samples, with lower median expression in both RA subtypes. Subtype-specific differences were also observed among proteins preferentially associated with ACPA-negative RA. MSMP showed lower expression in ACPA-negative RA compared with healthy controls and ACPA-positive RA, while CFHR5 exhibited intermediate expression levels across disease groups. Additional proteins, including PDCD1, NOS3, SERPINA9, PLA2G2A, TIMD4, TRIM39, RERG, and UBAP2, demonstrated moderate but consistent expression shifts across subdiagnoses. Overall, expression analysis of top-ranked proteins supported the model validity.

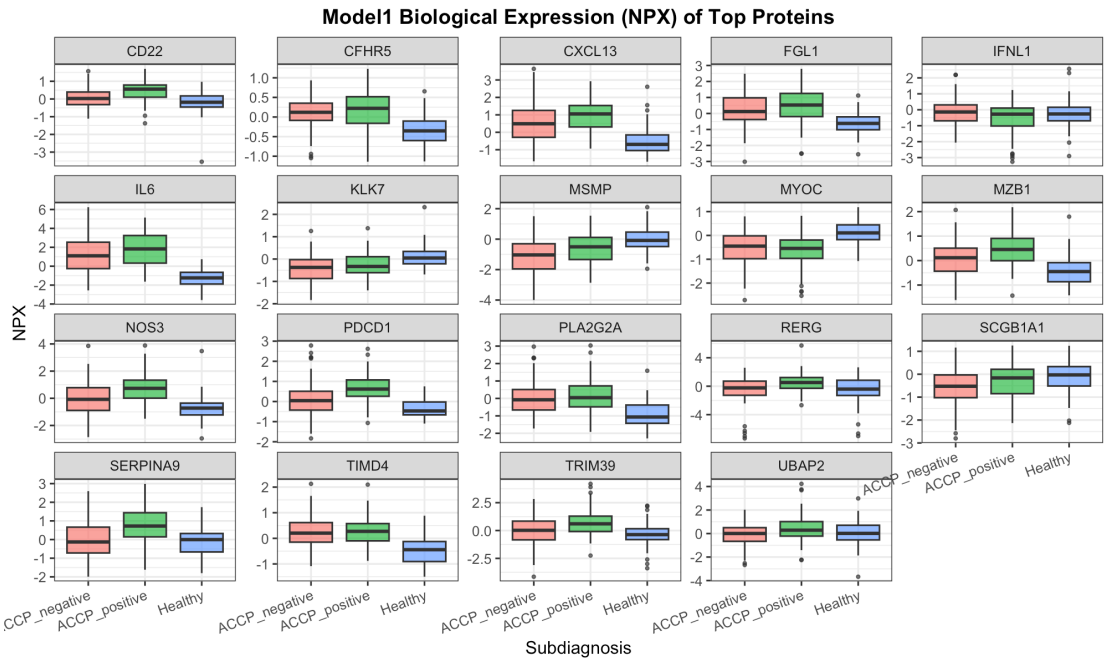

Fig. 3: Expression levels (NPX) of top selected proteins from *Model 1* compared across 3 groups (healthy controls, ACPA-positive and ACPA-negative).

Functional Enrichment Analysis of Top Selected Proteins

GO enrichment analysis was performed to test whether the top selected proteins from

Model 1 share any biological processes. Several immune-related pathways were significantly enriched (**Fig. 4**). The most strongly enriched term was regulation of cell activation (GO:0050865), with the highest gene ratio (11/41) and the lowest adjusted p-value. Contributing proteins included PDCD1, IL6, NOS3, CD22, KITLG, PLA2G2A, FGL1, MZB1, and PDCD1LG2, many of which were consistently selected by the classification models and showed subtype-specific expression patterns. The adaptive immune response (GO:0002250) pathway was also significantly enriched (9/41). This term involved proteins such as PDCD1, IL6, CXCL13, TNFRSF17, IL9, FGL1, PDCD1LG2, and SKAP1, which aligned with the strong contribution of lymphocyte-associated proteins observed in feature selection and expression analyses. Several innate immune-related pathways were enriched with moderate gene ratios, including cellular response to lipopolysaccharide (GO:0071222), defense response to bacterium (GO:0042742), and cellular response to molecule of bacterial origin (GO:0071219). These pathways shared overlapping gene sets, such as IL6, CXCL13, NOS2, NOS3, CASP1, and PLA2G2A, suggesting convergence on inflammatory and microbial response processes. In addition, integrin activation (GO:0033622) was enriched with a smaller gene set (CXCL13, MZB1, SKAP1), which indicated the involvement of cell adhesion-related signaling among a subset of model-selected proteins. Across enriched GO terms, gene ratios ranged from approximately 3/41 to 11/41, with adjusted p-values clustered within a narrow range.

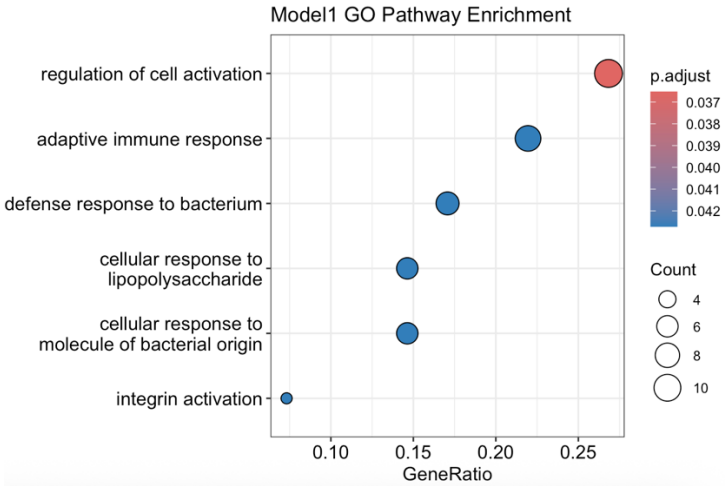

Fig. 4: Top GO terms selected by pathway enrichment analysis

Protein-Specific Interpretation

(i) PDCD1

PDCD1, encoding programmed cell death protein 1 (PD-1), was one of the most consistently selected features in the multiclass models and showed a strong association with ACPA-positive RA. PDCD1 is an immune checkpoint receptor expressed on T cells and B cells and maintains peripheral immune tolerance. In RA, PDCD1 is highly expressed on synovial T cells, reflecting persistent immune activation. Dysregulation of the PD-1–PD-L1 signaling axis results in insufficient downregulation of pathogenic immune responses. Particularly, PDCD1 is strongly expressed on peripheral helper T

cells in the synovium, where it supports B-cell activation and autoantibody production (Lowe et al. 2023; Tuttle et al. 2023). Therapeutically, PD-1 agonism has emerged as a potential strategy in RA. Peresolimab, a humanized IgG1 monoclonal antibody that binds to and activates PD-1, has progressed to phase II clinical trials and aims to restore inhibitory signaling in persistently activated synovial T cells (Lowe et al. 2023; Tuttle et al. 2023).

(ii) IL6

IL6 was a dominant contributor across binary and multiclass models and showed consistently elevated NPX levels in RA, with the highest expression observed in ACPA-positive disease. IL6 is a pleiotropic cytokine involved in immune regulation, tissue regeneration, and metabolic processes. In RA, dysregulated IL6 production contributes to chronic inflammation. Elevated levels of both IL6 and IL6 receptors were detected in serum and synovial fluid of affected joints. IL-6 and IL-6R-targeting therapies have demonstrated acceptable safety and efficacy profiles (Pandolfi et al. 2020; Jarlborg and Gabay 2022). At the pathway level, IL6 contributed to multiple enriched GO terms, including regulation of cell activation, adaptive immune response, and pathways related to cellular responses to bacterial or inflammatory stimuli. This supported its role as a shared driver across multiple immune processes rather than a subtype-restricted marker.

(iii) PADI4

PADI4 encodes protein-arginine deiminase type 4, an enzyme that catalyzes citrullination of arginine residues on histones and other proteins. It showed a subtype-specific association with ACPA-positive RA in the model. PADI4-mediated citrullination alters chromatin structure and gene regulation and plays a key role in neutrophil extracellular trap (NET) formation during innate immune responses. NET formation contributes to the generation of citrullinated autoantigens, directly linking PADI4 activity to ACPA production. Genetic studies have identified PADI4 as a susceptibility gene for RA across multiple populations, including Caucasian cohorts (Gandjbakhch et al. 2009; Suzuki et al. 2003; Worthington and John 2003). Several studies further demonstrate a stronger association between PADI4 variants and ACPA-positive RA, consistent with its selective contribution in this study (Massarenti et al. 2021).

(iv) CXCL13

CXCL13, C-X-C motif chemokine 13, was repeatedly selected across models and showed elevated expression in RA, particularly in ACPA-positive disease. CXCL13 is a key chemokine involved in positioning and activation of immune cells at lymphoid and ectopic lymphoid sites. CXCL13 binds to its receptor CXCR5 and attracts CXCR5-expressing B cells and follicular helper T cells. In RA, CXCL13 expression is increased in both early and established disease. TNF-α and IL6 upregulate CXCL13 expression within the synovium, promoting the formation of B-cell follicles and germinal center-like structures. The CXCL13-CXCR5 axis supports B-cell maturation into plasma cells and autoantibody production. Its consistent selection and expression gradient across

subdiagnoses align with the adaptive immune-dominant profile of ACPA-positive RA (Bechman et al. 2020).

(v) MSMP
MSMP, Prostate-associated microseminoprotein, showed preferential association with ACPA-negative RA in the model. MSMP acts as a ligand for the chemokine receptor CCR2 and induces chemotactic responses and intracellular calcium mobilization. HPA blood data indicated divergent expression patterns between RA and healthy controls, with directionality consistent with the model coefficients (healthy ~0.9; RA ~−1.3).

Discussion

This study applies machine learning models to large-scale plasma proteomics data to investigate molecular heterogeneity in RA. The results demonstrate that circulating protein profiles capture biologically meaningful signals that distinguish RA from healthy controls and further separate ACPA-positive and ACPA-negative subtypes. Proteins prioritized by elastic net models are not random statistical features but converge on immune activation, T-cell regulation, and B-cell-mediated processes related to RA pathogenesis. Several consistently selected proteins, including PDCD1, IL6, PADI4, and CXCL13, align with known subtype-specific mechanisms. ACPA-positive RA shows enrichment of adaptive immune and autoantibody-related pathways, whereas ACPA-negative RA exhibits a distinct and less well-characterized proteomic pattern. These findings support the heterogeneity of RA and highlight the value of proteomics-based stratification of disease subtypes.

This study has several limitations. First, sample size limits the power of subgroup analyses, particularly for sex-stratified models. Although sex-specific models suggest different proteomic patterns between female and male patients, the limited sample size compromises model stability and prevents reliable identification of sex-dependent protein signatures. As a result, the present analysis is unable to fully evaluate the contribution of age and sex to RA subtype classification. In addition, the study relies on cross-sectional data from a single cohort. Temporal dynamics of protein expression, treatment effects, and disease progression cannot be assessed. Longitudinal proteomic data would be required to determine whether the identified proteins reflect stable disease mechanisms, transient inflammatory states, or treatment-responsive signals.

Second, the analysis is restricted to plasma proteomic measurements, which limits the ability to infer tissue of origin and cell type information of the circulating proteins. Plasma reflects a composite signal from multiple tissues and immune compartments and does not capture spatial or cellular context within affected joints or secondary lymphoid organs. As a result, important tissue-specific processes, such as synovial fibroblast activation or localized immune cell interactions, cannot be directly resolved from plasma data alone. Incorporating tissue- and cell-level information will be necessary to achieve a more comprehensive understanding of RA pathogenesis.

Third, biological interpretation is limited to integration with existing databases and literature. While consistency with UniProt annotations, HPA expression patterns, and published studies supports the plausibility of selected proteins, no functional experiments are performed. Consequently, the analysis cannot establish causal roles for the identified proteins, nor can it directly validate their utility as diagnostic biomarkers or therapeutic targets. Translation to clinical application therefore remains challenging, as statistical association alone is insufficient for application in practice.

Finally, although machine learning models achieve good classification performance, the primary goal of this study is mechanistic insight rather than clinical prediction. In addition, the modeling strategy is limited to elastic net-regularized generalized linear models. While this approach provides interpretability and stable feature selection, alternative machine learning methods, such as random forest, are not explored and may capture complementary patterns.

Future work should focus on mechanistic interpretation and validation of the prioritized proteins. Beyond single-layer proteomic analysis, integrating circulating protein profiles with additional information, such as genomics, lipidomics and metabolomics data, may improve the ability to distinguish proteins that reflect core disease mechanisms from those representing downstream inflammatory effects. At the modeling level, further optimization of the analytical framework is needed. While elastic net models provide stable feature selection and interpretability, incorporating complementary modeling strategies may capture additional structure in high-dimensional proteomic data.

In addition, extending the current modeling framework to larger population-level datasets will be important to assess reproducibility and generalizability of the identified protein biomarkers. Linking circulating proteins to upstream genetic regulation or pathway-level context may further strengthen causal interpretation. Ultimately, moving from association-based protein selection toward mechanism-informed prioritization and validation, including integration of tissue-specific information and functional validation in animal models, remains a key step for translating proteomics-based findings in RA into biologically and clinically meaningful insights.

Overall, this study demonstrates that applying machine learning to high-dimensional plasma proteomics can identify biologically coherent and subtype-relevant protein biomarkers in RA. While the current findings are exploratory and hypothesis-generating, they offer a structured foundation for future mechanism-oriented studies aimed at improving RA subtype differentiation, supporting mechanistic investigation, and informing diagnosis and therapy.

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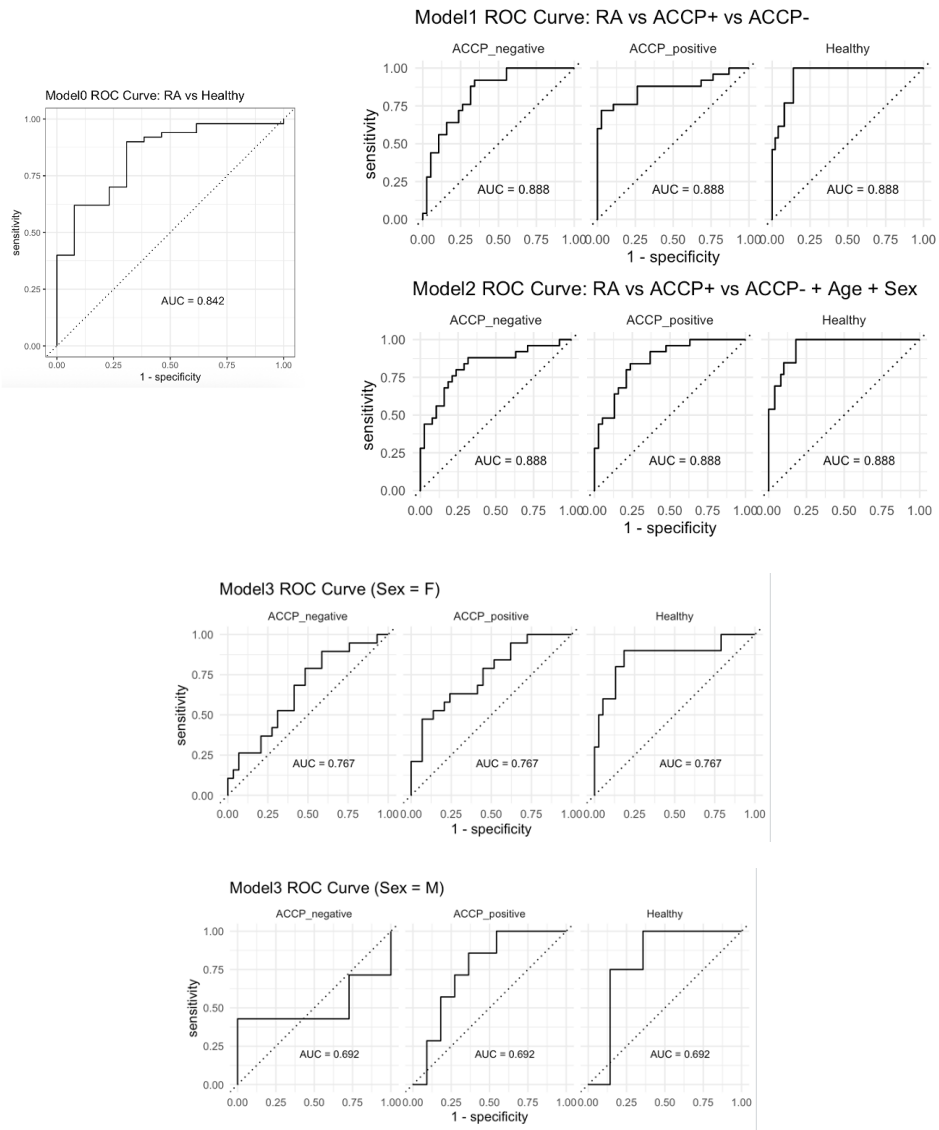
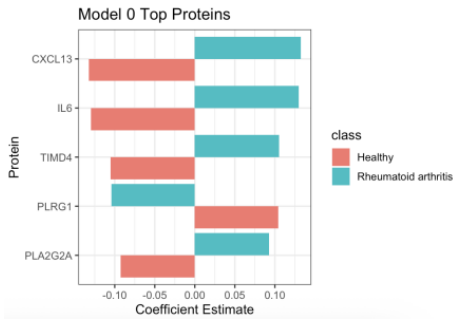
Appendix



Supplementary Fig.1: Model performances



Supplementary Fig. 2: Top-ranked proteins selected by *Model 0*

I used ChatGPT (https://chatgpt.com/) to gain a rapid overview over key concepts and correct for expression.